Save My Exams

GCSE · OCR · Computer Science

3 hours 69 questions

Data Storage & Compression

Data Storage & Compression

Units of Data Storage / Processing Binary Data / Data Capacity & Calculating Capacity Requirements / Converting Between Denary & Binary / Binary Addition / Converting Between Denary & Hexadecimal / Converting Between Binary & Hexadecimal / Binary Shifts / Representing Characters / Representing Images / Representing Sound / Compression

Easy (17 questions)	/43
Medium (26 questions)	/69
Hard (26 questions)	/79
Total Marks	**/191**

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Easy Questions

1 (a) Computers represent data in binary form.

Tick (✓) **one** box **in each row** to identify the **binary unit equivalent** of each of the given file sizes.

File size	2 megabytes	2 petabytes	2 kilobytes	2 bytes	2 gigabytes
2000 bytes					
2000 terabytes					
16 bits					
4 nibbles					

(4 marks)

(b) **Convert** the **denary number 221** into **8 bit binary**.

Show your working.

.....

..... **(2 marks)**

(c) **Convert** the **hexadecimal number 2F** into **denary**. Show your working.

..... **(1 mark)**

(d) **Convert** the **binary number 10110000** into **hexadecimal**

..... **(1 mark)**

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(e) **Identify** how many **unique values** can be represented by **4 bits**.

..... **(1 mark)**

(f) **Perform** a **binary shift** of **3 places right** on the binary number **10001110**.

..... **(1 mark)**

2 (a) The student writes a report about volcanoes.

The computer stores text using the **ASCII character set**.

Part of the ASCII character set is shown:

Character	ASCII denary code
M	77
N	78
O	79
P	80
Q	81

Identify the **character** that will be represented by the **ASCII denary code 84**.

..... **(1 mark)**

(b) **Identify** a second **character set**.

..... **(1 mark)**

3 A student takes a photograph of their science experiment. The image file includes **metadata**.

Identify three pieces of **metadata** that is often stored with an image.

.....

.....

..... **(3 marks)**

4 **Convert** the **binary number** 11001011 into **denary**.

..... **(1 mark)**

5 (a) **ASCII**, **extended ASCII** and **Unicode** are all examples of character sets.

Tick (✓) one or more boxes in each row to identify whether each statement applies to each character set.

	ASCII	Extended ASCII	Unicode
Can represent thousands of different characters, including Russian and Chinese symbols.			
Can represent European characters such as ç or â.			
Uses different character codes for upper-case and lower-case letters			

(3 marks)

(b) The **character D** is represented by the binary **ASCII code 1000100**.

Give the ASCII code for the following characters **in binary**.

E

H

.....

..... **(2 marks)**

6 **Convert** the **denary** value **178** into an **8 bit binary** number.

.....

..... **(2 marks)**

7 **Convert** the **binary** value **11000111** into **hexadecimal**.

.....

..... **(2 marks)**

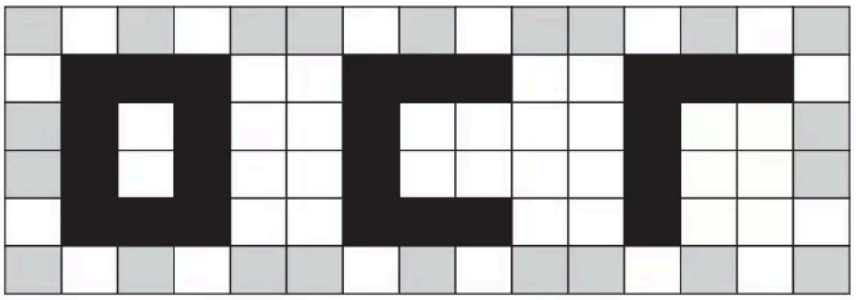
8 **Convert** the **denary** value **105** into an **8 bit binary** number

.....

..... **(2 marks)**

9 The following logo is stored as a **bitmap image**.

Each box represents **one pixel**, with three different colours being used in the image.



State what is meant by the term **image resolution**.

..... **(1 mark)**

10 (a) **Give two ways** that the **file size** of an image could be **reduced**.

.....

..... **(2 marks)**

(b) **Metadata** is sometimes stored alongside images.

State what is meant by the term **metadata**.

.....**(1 mark)**

(c) **Give one example** of **metadata** that could be stored alongside the logo.

.....**(1 mark)**

11 **Convert** the **binary number** 1011011 to **denary**.

Show your working.

.....

.....**(2 marks)**

12 **Convert** the **denary** number **132** into an **8 bit binary** number.

.....

.....**(2 marks)**

13 **Convert** the **binary** number **10110101** to its **hexadecimal** equivalent.

.....

.....**(2 marks)**

14 **Show** the result of a left binary shift of 4 places on the binary number 0000 1111.

.....**(1 mark)**

15 A musician uses a computer to make and record music.

Tick (✓) one box to identify the correct description of sound sampling.

A. The frequency of the wave is measured a set number of times each second.

B. The amplitude of the wave is measured at set intervals.

C. The digital sound wave is measured a set number of times each second.

D. The analogue sound wave's resolution is measured at set intervals.

(1 mark)

16 **Tick (✓)** one box to identify the smallest secondary storage capacity.

A. 2.1 GB

B. 300 MB

C. 200 000 KB

D. 0.0021 TB

(1 mark)

17 **Convert** the **hexadecimal** value **A5** into **binary**.

.....

..... **(2 marks)**

Medium Questions

1 (a) A student **compresses** all their documents before emailing them to their teacher

Give two benefits of **compressing the data** before it is emailed.

.....

..... **(2 marks)**

(b) **Explain** why **lossy compression** may **not be appropriate** to compress all of the student's files.

.....

..... **(2 marks)**

2 The **sample rate is increased** on a computer when recording music from a guitar.

Give two effects this will have on the recording.

.....

..... **(2 marks)**

3 (a) **Complete** a **2-place shift to the right** on the binary number 11001011.

..... **(1 mark)**

(b) **Explain the effect** of performing a **2-place shift to the right** on the binary number 11001011.

..... **(1 mark)**

4 The table gives the **ASCII code** for the characters.

Character	ASCII code
L	76
M	77
N	78
O	79
P	80

Explain how the word **MOP** will be represented in **ASCII**.

.....

..... **(2 marks)**

5 Computers make use of electronic switches called **transistors**.

Describe how transistors can be used to **store a value in binary**.

.....

..... **(2 marks)**

6 (a) Azmi says, "hexadecimal is used because it takes up less storage space in the computer's memory than binary."

Tick one box to identify whether Azmi is correct.

Justify your answer.

	Tick (✓)
Correct	
Incorrect	

Justification	

.....

..... **(2 marks)**

(b) **Binary shifts** can be used for **multiplication** and **division**.

Draw one line from each shift on the left to its correct outcome on the right.

Draw **one** line from each shift on the left to its correct outcome on the right.

Binary shift	**Outcome**
Right shift of 2 places on 1010 1000	0011 1010, divides by 4 with a loss of precision
Left shift of 1 place on 0010 1101	0010 1010, divides by 4
Right shift of 2 places on 1110 1001	0101 1010, multiplies by 2
Left shift of 3 places on 0001 1111	1111 1000, multiplies by 8

(3 marks)

(c) **Add** the following **8 bit binary integers**, giving your answer in **binary**.

```
  0 0 1 1 0 1 1 0
+ 1 0 0 1 0 1 1 0
-----------------
```

.....

..... **(2 marks)**

7 Ali's tablet computer has **100GB of secondary storage**. There is currently **80GB available**.

Ali wants to transfer a series of video clips onto his tablet. **Each video** is, on average, **200000 kilobytes**.

Calculate an estimate of the **number of video clips** Ali can fit onto his tablet.

Show your working.

.....

.....

.....

..... **(4 marks)**

8 **Give two reasons** why computer scientists **use hexadecimal to represent numbers** instead of binary.

.....

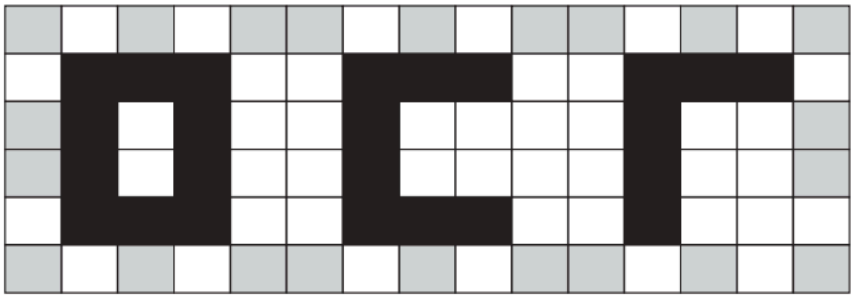
..... **(2 marks)**

9 **Show** the outcome of a **right shift of three places** on the binary value **0111 1000**.

..... **(1 mark)**

10 The following logo is stored as a **bitmap image**.

Each box represents **one pixel**, with three different colours being used in the image.



Calculate the fewest **number of bits** that could be used to **store the logo as a bitmap image**.

You must **show your working**.

.....

.....

.....

..... **(4 marks)**

11 A radio station uses a digital camera to take a photograph of a computer scientist for their website. The photograph is stored as a **bitmap image**.

Describe how bitmap images are represented in binary.

.....

.....

..... **(3 marks)**

12 **Explain why** computers **represent data in binary form**.

.....

..... **(2 marks)**

13 An image is **compressed using lossy compression** before being uploaded to a radio station's web server.

The image will be used on the radio station's website.

Describe one **advantage** and one **disadvantage** of **using lossy compression** on the image that will be used on the website.

.....

.....

.....

..... **(4 marks)**

14 **Convert the hexadecimal** number **A3** to **denary**.

Show your working.

.....

..... **(2 marks)**

15 **Add** the following binary numbers.

```
  10110110
+   100111
----------
```

.....

..... **(2 marks)**

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16 William transfers videos to a computer for editing.

The computer has **1GB** of **storage free**.

Calculate the **number of videos** that could be stored on the computer if each video was **100MB** in size.

Show your **working**.

.....

..... **(2 marks)**

17 **Show the effect** of a binary **shift right of two places** on the binary number **00110100**.

..... **(1 mark)**

18 **Describe** a **shift** that can be used to **double the value** of the binary number **00100100**.

.....

..... **(2 marks)**

19 The table below (Fig. 3) shows the **ASCII codes** for a number of characters.

The lower case ASCII code for a character can be found by **adding 00100000** to the upper case version.

Character	ASCII code
R	0 1 0 1 0 0 1 0
r	0 1 1 1 0 0 1 0
A	0 1 0 0 0 0 0 1
a	
E	0 1 0 0 0 1 0 1
e	

Fig. 3

Complete the table above by filling in the **missing ASCII codes**.

(2 marks)

20 (a) The following table has either the binary or denary value of 3 numbers.

Complete the table by converting the 8-bit binary number into denary and the denary number into 8-bit binary.

8-bit Binary	Denary
11110000	
	105
00011110	

(3 marks)

(b) **Complete** the table by writing the answer to each statement

Statement	Answer
The smallest denary number that can be represented by a 4-bit binary number	
The largest denary number that can be represented by a 6-bit binary number	
The maximum number of different colours that can be represented with a colour depth of 7-bits	
The minimum number of bits needed to represent 150 different characters in a character set	

(4 marks)

21 **Add** these two 8-bit binary numbers using binary addition.

Show your working out.

$$\begin{array}{r} 0\,1\,1\,0\,1\,0\,1\,1 \\ +\,0\,0\,0\,0\,1\,1\,1\,1 \\ \hline \end{array}$$

.....

..... **(2 marks)**

22 **Explain** how changing the bit depth will affect a sound file.

.....

..... **(2 marks)**

23 The musician's recordings have an average (mean) file size of 3 MB. The musician has 1000 recordings.

Calculate an estimate of the storage space in GB that the 1000 files will require, assuming they are each 3 MB in size.

Show your working out.

.....

..... **(2 marks)**

24 A software developer is creating a mobile application. Each image file used in the application has an average (mean) file size of 2 MB. The application contains 500 image files.

Calculate an estimate of the storage space in GB that the 500 files will require, assuming they are each 2 MB in size.

Show your working out.

.....

..... **(2 marks)**

25 **Add** the following binary numbers.

```
  1 1 0 0 1 0 1 0
+       0 1 1 0 1 1
-----------------
```

.....

..... **(2 marks)**

26 **Add** the following binary numbers.

```
  1 1 1 0 1 0 1 1
+ 1 0 0 1 1 1 0 1
-----------------
```

.....

..... **(2 marks)**

Hard Questions

1 (a) A student is creating a range of documents for a school project.

The student records a podcast about computer science.

Describe how an **analogue sound wave** is **converted into digital form**.

.....

.....

..... **(3 marks)**

(b) **Tick (✓) one or more** boxes on **each row** to identify the effect(s) that each change will have on the sound file.

Change	File size increases	File size decreases	Accuracy increases	Accuracy decreases
Duration changes from 10 minutes to 20 minutes				
Sample rate changes from 44 kilohertz to 8 kilohertz				
Bit depth changes from 8 bits to 16 bits				

(3 marks)

2 A computer records an audio file of someone playing a guitar.

Describe what happens when the computer **converts the music into a file**.

.....

..... **(2 marks)**

3 (a) Sound data is sampled and stored in binary.

A 30-second section of sound data is sampled at a rate of 48KHz using 24 bits per sample.

Describe the data that is recorded **when sound is sampled**.

.....

..... **(2 marks)**

(b) **Explain** what is meant by a **sample rate of 48KHz.**

.....

..... **(2 marks)**

(c) **Describe** how the **file size of the sound recording** could be **reduced**.

.....

.....

.....

..... **(4 marks)**

4 **Describe** how to convert a 2-digit hexadecimal number into denary.

Use an example in your answer.

.....

..... **(2 marks)**

5 **Calculate** the 8-bit binary representation for the denary number **253**

.....

..... **(2 marks)**

6 **Calculate** the 8-bit binary representation for the denary number **189**.

.....

..... **(2 marks)**

7 A network packet contains the 8-bit binary value **11010111**.

Calculate the denary equivalent and **state** whether the result is odd or even.

.....

..... **(2 marks)**

8 A sensor reading is stored as the 8-bit binary value **10110101**.

Calculate the denary equivalent and **state** whether the result is odd or even.

.....

..... **(2 marks)**

9 **Convert** the denary number **189** into a 2-digit hexadecimal value.

Show your working.

.....

.....

..... **(3 marks)**

10 **Convert** the denary number **227** into a 2-digit hexadecimal value.

Show your working.

.....

.....

..... **(3 marks)**

11 A colour code is stored as the hexadecimal value **E7**.

Calculate the denary value of this colour code.

Show your working.

.....

.....

..... **(3 marks)**

12 A memory address is stored as the hexadecimal value **B4**.

Calculate the denary value of this memory address.

Show your working.

.....

.....

..... **(3 marks)**

13 A MAC address segment is represented by the binary string **11101101**.

Convert this value into a 2-digit hexadecimal number.

Show your working.

.....

..... **(2 marks)**

14 A network interface card uses the binary value **10110011** as part of its unique identifier.

Convert this binary value into a 2-digit hexadecimal number.

Show your working.

.....

..... **(2 marks)**

15 **Convert** the hexadecimal value **A9** into an 8-bit binary number.

Show your working.

.....

..... **(2 marks)**

16 **Convert** the hexadecimal value **F3** into an 8-bit binary number.

Show your working.

.....

..... **(2 marks)**

17 Perform a **3-place left shift** on the binary number **00010100**.

Evaluate the effect this has on the original denary value.

.....

.....

..... **(3 marks)**

18 Perform a **4-place right shift** on the binary number **11010000**.

Evaluate the effect this has on the original denary value.

.....

.....

..... **(3 marks)**

19 A developer is creating a custom character set to represent **500 unique symbols**.

Explain the minimum number of bits required to store each character and **evaluate** the impact this would have on file size compared to using **7-bit ASCII**.

.....

.....

..... **(3 marks)**

20 In an 8-bit Extended ASCII system, the character **A** has the denary value **65**.

Calculate the binary representation of the character **J** and **explain** why an 8-bit system is needed for modern symbols such as the copyright sign (©).

.....

.....

..... **(3 marks)**

21 A global social media platform moves from **8-bit Extended ASCII** to **16-bit Unicode**.

Analyse the benefit of this change for international communication and evaluate the impact on secondary storage.

.....

.....

.....

..... **(4 marks)**

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22 An image's colour depth increases from **8 bits to 24 bits**, and its resolution doubles in both height and width.

Calculate the factor by which the file size increases and **analyse** the effect on image quality.

.....

.....

.....

..... **(4 marks)**

23 **Identify** which items of metadata are required to correctly display a bitmap image and **explain** how metadata helps manage large photo libraries.

.....

.....

..... **(3 marks)**

24 A video-on-demand service streams **4K video** to mobile users.

Evaluate the need for compression in this scenario.

.....

.....

.....

..... **(4 marks)**

25 A musician emails a high-quality audio recording to a producer for a quick review.

Justify the use of **lossy compression** for this task.

.....

.....

..... **(3 marks)**

26 A business archives its legal financial contracts digitally.

Analyse why **lossless compression** is the only suitable option.

.....

.....

..... **(3 marks)**